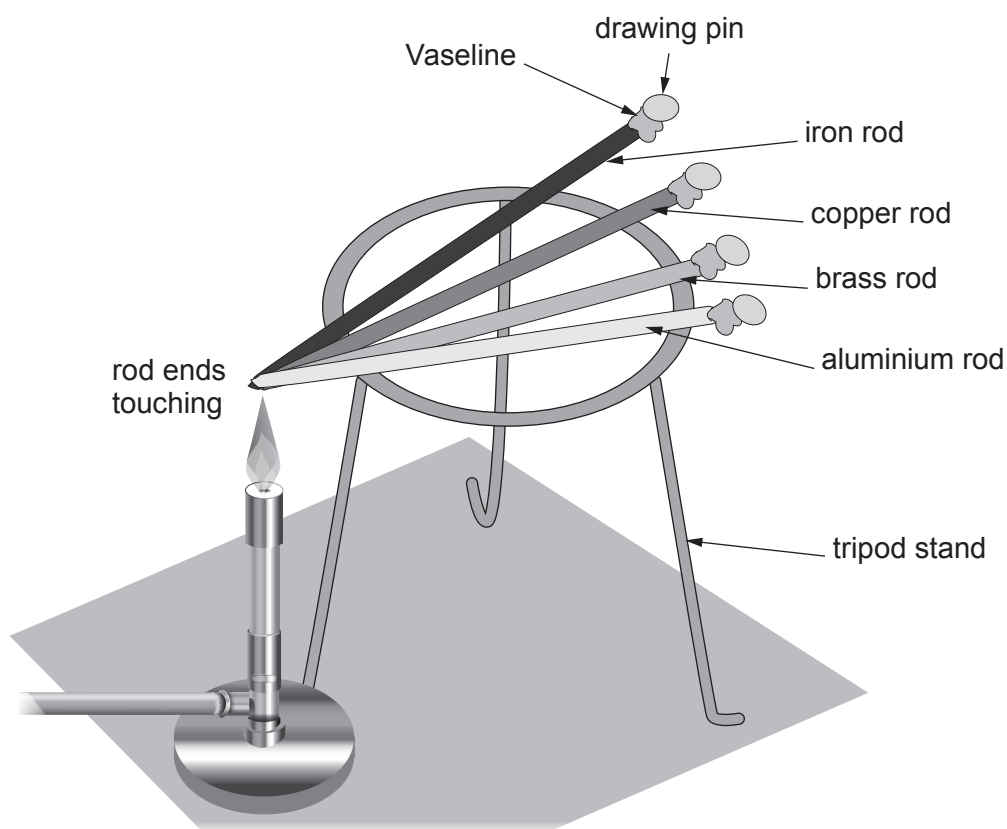


4. (a) (i) The apparatus shown can be used to compare how well different metals conduct heat.

Explain:

- how the apparatus is used
- how the results are used to show that copper is the best conductor.

[3]



- (ii) Describe, in terms of the behaviour of **particles**, how heat energy is conducted through the metal rods.

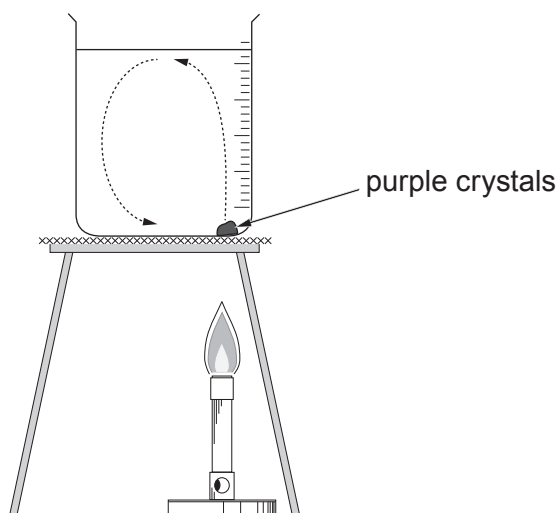
[2]



- (b) A teacher demonstrates the process of convection in liquids.

Some purple crystals are placed in the bottom corner of a beaker that is filled with water.

The water is then heated and a current of coloured water is seen to move as shown in the diagram below.



- (i) Explain why the coloured water moves in the way shown above.

[2]

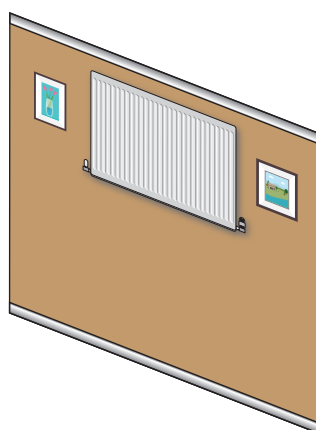
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- (ii) Explain why it is not suitable to put a hot radiator at the top of a wall to warm the room.

[2]



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